

Computer Vision: Python Setup

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We will be using Python (specifically python3) for all the assignments and breakout sessions in this class. Historically, the setup part has been somewhat tricky to get right, but recent progress in Python package managers has made this much easier. We will be using a tool called “uv” for this purpose. It’s brilliant.

I generally recommend that you use either MacOS or Linux for these assignments. If you are experienced with Python on Windows, that’s no problem, but I can only provide limited help if you become stuck, as I do not have a Windows machine for development. If you have a Windows computer, I recommend you look into WSL, the Windows Subsystem for Linux, which lets you run a headless Linux environment from within Windows.

Getting Started

I recommend that you save all assignments into an ‘assignment parent folder.’ Copy-paste the provided pyproject.toml file into a parent folder, from which you can run the uv commands. The uv package manager handles virtual environment setup for you. Install it, move to the correct folder (via the cd “change directory” command) and then run the notebook. Go to their website and install it: <https://docs.astral.sh/uv/getting-started/installation/#standalone-installer> Then do the following:

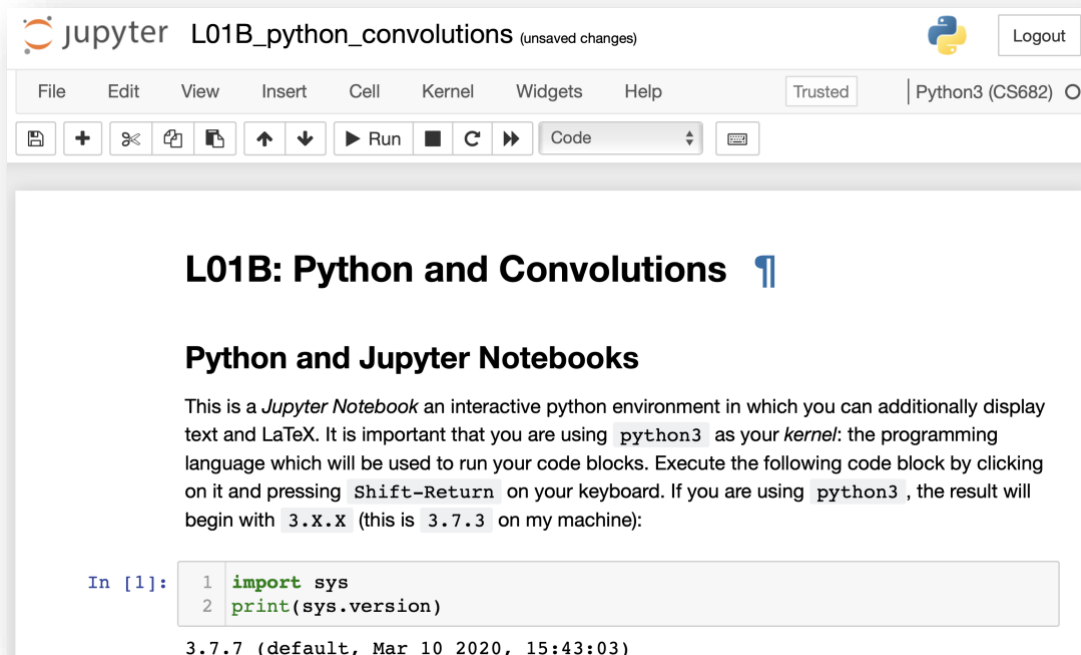
```
# Move to this folder
cd $ASSIGNMENT_PARENT_FOLDER

# Create a virtual environment and activate it
uv run launch-assignment-notebooks

# This is equivalent to the above:
uv run jupyter notebook
```

Configure and run the Jupyter Notebook

We will be using a Jupyter Notebook (formerly iPython) for running code in this class. It provides an excellent way to write snippets of code, create figures, and include helpful information and LaTeX-style equations all in a single document. Upon running it, you should see a screen that looks like this:



Once you have opened the notebook, you can follow the instructions there to move forward.